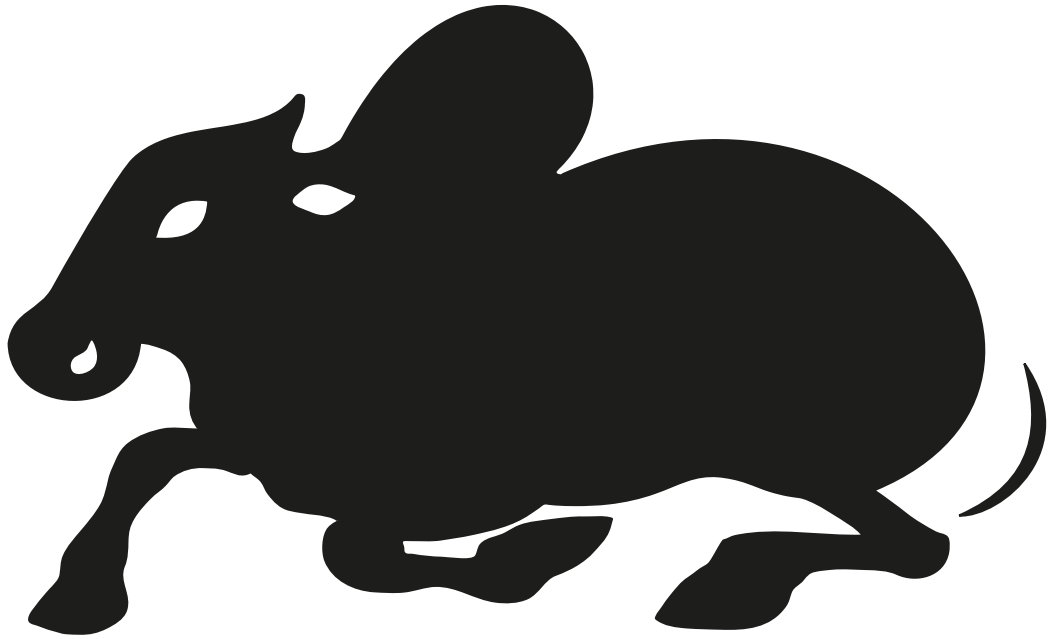

dharma

Versification Guide

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Contents

1. Introduction **2**

1.1. About ... 2
1.2. Versification terminology ... 2
1.3. Syllable length ... 3
1.4. Prosodic code ... 4
1.4.1. How to use prosodic code ... 4
1.4.2. When to use prosodic code ... 4
1.5. *Gaṇa* notation ... 5

2. The Sanskritic versification tradition **7**

2.1. Identifying Sanskritic metres ... 7
2.2. Syllabic metres (*varṇavṛtta*) ... 9
2.2.1. *Anuṣṭubh* ... 9
2.2.2. The *upajāti* group ... 10
2.2.3. Pre-classical trimeter ... 10
2.2.4. The *daṇḍaka* class ... 11
2.3. Moraic metres ... 11
2.3.1. The structure of moraic lines ... 12
2.3.2. The structure and classification of moraic stanzas ... 13
2.4. Metres combining syllabic and moraic cola ... 14
2.4.1. The *vaitālīya* group ... 14
2.4.2. *Gītikā* ... 15

3. The Tamil versification tradition **16**

4. Bibliography **17**

1. Introduction

1.1. About

This is the first version of the standalone DHARMA Versification Guide (VG), which is a revised version of the former Appendix B of the DHARMA Encoding Guide for Diplomatic Editions. Because its relevance is broader than that specific Guide, it has been reworked into a separate document.

1.2. Versification terminology

- **verse**: used as an uncountable noun to refer to text characterised by rhythmically iterated units (generally prosodic units in our case)
  - to avoid ambiguity, DHARMA guides never use "verse" as a countable noun meaning either "stanza" or "line" (as defined below), although both are legitimate meanings of this word in general usage
- **stanza**: a unit of verse characterised by a (usually prosodic) pattern and consisting of a number of smaller units (lines) that do not, as a rule, occur in less than a full stanza
  - in Sanskritic syllabo-quantitative verse, *stanza* is usually equivalent to *catuṣpadī*
    - the term ***quatrain*** may be used as a synonym for a stanza consisting of four lines
  - in Tamil verse, a *stanza* in the usage of DHARMA guides is equivalent to a "poem" (*pā*, *pāṭṭu*, *ceyyuḷ*)
- **line**: a unit of verse characterised by a (usually prosodic) pattern, with several lines (which may have identical or different patterns) making up a stanza
  - in Sanskritic syllabic verse, *line* is equivalent to *pāda*
  - in Sanskritic moraic verse, *line* is for our purposes equivalent to *hemistich*
  - in Tamil verse, *line* is equivalent to *aṭi*
- **quarter** is used as a synonym of *line* in the context of Sanskritic syllabo-quantitative verse in order to reduce ambiguity by clearly distinguishing verse lines (an element of intrinsic structure) from physical lines (an element of extrinsic structure)
- **hemistich**: a half-stanza, the first or the second pair of lines in a quatrain
  - note that the term *hemistich* originates from European classical prosody where it is used in a different sense (a half-line), but the term has been widely applied in European discussions of Indic prosody in the sense in which we use it here
- **break** or line break: a boundary between lines or hemistichs, which usually coincides with a word boundary
  - **enjambement** in our usage means the occurrence of a line break within a word (usually between members of a compound; rarely within a morpheme)
  - note a metrical line break is not the same as a break in physical lines (EGD §4.5), though in modern typography a physical line break often coincides with a metrical line break
- **caesura**: a boundary within a line, which divides the line into smaller prosodic units (*cola*) and as a rule coincides with a word boundary
- **colon** (plural *cola*): a prosodic unit smaller than a line, a division resulting from the presence of a caesura
- **foot**: a small prosodic unit that has no regard for word boundaries
  - in the present Guide this term is only used as applicable to Sanskritic moraic verse, where feet (*gaṇa*) consist of a set number of morae
- **mora** (plural *morae*; Sanskrit *mātrā*): a unit of prosodic length defined as the length of a short syllable
  - in the Sanskritic tradition, the length of all long syllables is counted as two morae
- **metre**: a prosodic template (a fixed pattern of syllables or feet), which has a conventional name
- **scansion** refers to the process of establishing the prosodic pattern of a segment of text, as well as to the established prosodic pattern
- **syllabic metre** in our usage refers to all metres defined on the basis of a (largely) fixed number of syllables and constraints applicable to some or all of these syllables

  - this corresponds to *varṇavṛtta* in the Sanskritic tradition
  - in stricter usage, which is not relevant here, syllabic metre refers to metres constrained only by the number of syllables, while the metres we deal with are predominantly syllabo-quantitative (i.e. based on both syllable count and quantity constraints)
- an **anceps** syllable is a syllable of a syllabic metre which may be either long or short (as discussed further in §1.3)
- **moraic metre** in our usage refers to metres defined on the basis of feet or cola constituted of a fixed number of morae but with little or no constraint on the number of the syllables comprising them
  - this corresponds to *mātrāvṛtta* in the Sanskritic tradition, especially to the foot-based *gaṇacchandas* category

1.3. Syllable length

Versification in the Sanskritic tradition as well as in several other languages we work with usually relies on the distinction of short and long syllables for producing rhythm in verse, therefore we prefer the terms "length", "long" and "short" to the corresponding triad "weight" (or "quantity"), "heavy" and "light" often used in discussions of verse in other languages. In this context, "syllable" always refers to a prosodic syllable, which is not necessarily identical to a linguistic syllable, nor to the *akṣara* of Indic writing systems.

- a prosodic syllable is a segment starting with a vowel (or with one or more consonants followed by a vowel when at the beginning of a metrical unit) and ending before the next vowel or at the end of a metrical unit, regardless of word boundaries
  - the *anuṣṭubh* hemistich *namas tuṅga-śiraś-cumbi-candra-cāmara-cārave* is thus broken down into prosodic syllables as follows: *nam-ast-uṅg-aś-ir-aśc-umb-ic-andr-ac-ām-ar-ac-ār-av-e*
  - note that many discussions of (especially Western) prosody demarcate prosodic syllables so that they begin with a consonant when one is present (e.g. *na-mas-tuṅ-ga-śi-raś-cum-bi*, etc.)
    - this yields an intuitively easier syllabification but makes the determination of syllable length less straightforward
- syllable length is determined as follows:
  - a **short syllable** is one whose vowel is short (in Sanskrit: *a*, *i*, *u*, *ṛ*, *ḷ*) and is followed by no more than one consonant (without regard to whether a word boundary is also present)
    - keep in mind that an aspirated consonant is one consonant even though it is transliterated by two Latin letters
  - a **long syllable** is one that does not meet both of the above conditions for a short syllable; specifically, a syllable is called
    - long by nature, if its vowel is long (in Sanskrit: *ā*, *ī*, *ū*, *ṝ*, *e*, *ai*, *o*, *au*)
    - long by position, if its vowel is followed by two or more consonants
- in the Sanskritic tradition, the last syllable of a metrical unit is conventionally regarded as long even if it is technically short
  - this convention is called *brevis in longo* ('short in [place of] long') in the European versification tradition
  - a syllable whose length is not governed by a prosodic constraint is called *anceps*)
  - in the Sanskritic tradition, this convention applies
    - in particular to the ends of hemistichs (and thus also of full stanzas)
    - often, but not necessarily, to the ends of lines that do not coincide with hemistich ends
    - and normally does not apply to the ends of cola
- *anusvāra* and *visarga* normally count as consonants in determining syllable length, but
  - in Prakrit languages, *anunāsika* (usually not distinguished in writing from *anusvāra*) indicates the nasalisation of a vowel rather than a nasal consonant following the vowel, and a nasalised vowel followed by a single consonant may still count as short
  - in Sanskrit verse (though generally only in pre-classical Sanskrit), *anusvāra* and *visarga* may cause the preceding vowel to be long even when followed by another vowel

- note that this normally entails no complication, as by standard sandhi, *anusvāra* or *visarga* will only occur before a consonant or at the end of a hemistich, in both of which cases the syllable containing the *anusvāra* or *visarga* will be long or count as long

1.4. Prosodic code

Classical prosodic notation for syllable length is based on breves ⏑ for short syllables and dashes – for long syllables. Because some of the relevant symbols are difficult to produce on a keyboard and have poor font support, in XML encoding we use a set of simple ASCII characters for recording prosody. Table 1 below shows the classical symbols relevant to our work (with which you should also familiarise yourself) and the corresponding ASCII characters to be used in encoding, along with the encoding contexts in which the various codes are relevant, discussed further in §1.4.2 below.

Table 1. Prosodic notation

Description	Classical notation	ASCII notation	Context
one short/light syllable	⏑	-	1, 2, 3
one long/heavy syllable	–	+	1, 2, 3
one syllable of indeterminate length (anceps)	⏒	=	1, 2, (3)
two morae (one long or two short syllables)	⏑⏑	2	1, 2, (3)
larger moraic foot or colon		numeral(s)	1, 2, (3)
foot boundary	|	|	1, 2, (3)
caesura	||	||[1]	1
boundary of odd and even quarter		/	1

1.4.1. How to use prosodic code

- prosodic code must never contain spaces
- when numbers are used to encode the length of moraic feet (or moraic cola), the number(s) representing the mora count of a foot or colon must always be separated
  - by the foot boundary sign | from adjacent moraic feet or syllabo-quantitative segments
  - by the caesura sign || from adjacent (moraic or syllabo-quantitative) cola
  - neither of these signs must be used at the end of a segment of metrical notation, i.e.
    - after the last foot or colon in a template
    - after the last (complete or partial) foot within a metrical lacuna
- using these separators between numbers makes the notation easier to scan and affords the possibility of using more than one digit to represent the mora count of a colon, should this be necessary
  - however, consider whether large moraic units can be analysed into combinations of smaller feet

1.4.2. When to use prosodic code

DHARMA editions may call for the encoding of prosody in the following contexts:

1. `@met` used in `<lg>` (or, rarely, `<l>`) to encode the metre for which a conventional name is not available (EGD §3.3.3.1)

[1] Two iterations of | (`U+007C` Vertical Line), not a || double vertical bar character.

2. `@met` used in `<seg>` to encode the prosody of a lacuna (EGD §6.5.2.3)
3. `@real` used in `<l>` to encode the actual prosody of a metrically deviant verse line (EGD §3.3.2 and §3.3.4.1)

The range of applicable notation differs in the above contexts, as indicated in the last column of Table 1 and elaborated below.

- in the attribute `@real` (context 3), use only the signs + and - to encode the exact prosody of a metrical realisation
  - except when a metrical anomaly is complicated by a (partial) lacuna in the anomalous segment, as a result of which the exact prosody of the segment cannot be determined, in which case you may use
    - = for anceps when the length of a syllable cannot be determined
    - numerals for moraic notation when the number of lost syllables cannot be determined but the expected mora count can be estimated
- in the attribute `@met` (contexts 1 and 2), use the notation for anceps (rather than that for a long syllable) at the end of each line of syllabic verse
  - caesurae shall only be encoded in context 1
  - odd/even quarter boundaries shall only be noted in context 1 and only when the metre involves a different template for odd and even lines (*ardhasamavṛtta*)

1.5. *Gaṇa* notation

The Sanskritic tradition represents scansion by a code consisting of a set of *akṣara*s that conventionally stand for sets of three prosodic syllables (*trika-gaṇa*) or for a single short or long syllable, as summarised in Table 2. These units are called *gaṇa*, the standard term for feet in moraic verse, but are not in fact feet in the sense of autonomous constituent units of larger prosodic patterns. This *gaṇa*-based notation is included in our authority file for prosodic patterns because it is often familiar to those brought up in the Indic tradition of versification, and is useful for looking up obscure metres in traditional treatises on poesy. However, we consider it on the whole a less useful tool for describing scansion, because with this notation the same metrical segment takes different forms depending on the number and length of syllables preceding it. Familiarity with this system is not required for working on DHARMA digital editions.

Where the *gaṇa* notation shines, on the other hand, is its efficacy in constructing learnable definitions for specific metres. Such definitions in the Sanskrit tradition generally contain the prosodic formula for a metre described using the *akṣara*s of this notation (or just their consonants), combined with logograms (*bhūtasaṁkhyā*) indicating the length of successive cola in metres involving one or more caesuras, containing the name of the metre, the whole definition itself also being set to the metre. For instance, a classical definition of the *śārdūlavikrīḍita* metre (– – – ⏑ – ⏑ – ⏑ ⏑ ⏑ – || – – ⏑ – – – ⏑ –) runs as follows: *sūryāśvair masajāḥ statāḥ saguravaḥ śārdūlavikrīḍitam* (*Vṛttaratnākara* 3.96). Here, *sūrya* and *aśva* are numerical codes for 12 (the number of Ādityas) and 7 (the number of the Sun God's horses), the length of the metre's two cola. The scansion is given through *m-s-j-s-t-t+guru*.

There are also a number of conventional memory aids for learning the notation itself. One is the (meaningless) string *yamātārājabhānasalagā*, in which the three syllables starting from any of the consonants (or just one syllable starting from one of the last two consonants) give the prosodic pattern denoted by that consonant, as shown in the third column of Table 2. Another is a set of (meaningful but irrelevant) words beginning with each of the consonants, which likewise show the pattern associated with the consonant, as shown in the last column of Table 2.

Table 2. Gaṇa notation

code	scansion	string segment	word
ya	⏑ – –	yamātā…	yathārtham
ma	– – –	mātārā…	mantrāṇām
ta	– – ⏑	tārāja…	tantrāṇi
ra	– ⏑ –	rājabhā…	rakṣaṇam
ja	⏑ – ⏑	jabhāna…	jajāpa
bha	– ⏑ ⏑	bhānasa…	bhaṅguram
na	⏑ ⏑ ⏑	nasala…	nayati
sa	⏑ ⏑ –	salagā	sahajam
la	⏑	la	(laghu)
ga	–	gā	(guru)

2. The Sanskritic versification tradition

In the umbrella term "Sanskritic versification tradition", we include the versification traditions of Middle Indo-Aryan (Prakrit) languages which may have roots outside the school of Sanskrit versification but are nonetheless inextricably intertwined with the latter in Indic poetry. We likewise include here the Sanskrit-derived traditions associated with Dravidian and Southeast Asian languages.

2.1. Identifying Sanskritic metres

Although this may be difficult for native speakers of languages that do not have a concept of syllable length (as opposed to weight), we strongly recommend that you try reciting the verses you encounter and timing the syllables correctly. This will help tremendously in recognising metres. For any verse whose metre you cannot recognise confidently, we recommend the following.

- transcribe the prosody of the stanza (or at least one hemistich) using either the classical or the ASCII notation
  - if the lines have an identical scansion or differ only to a small extent, you are probably dealing with a syllabo-quantitative metre
  - if the scansion of the lines differs greatly, you are dealing with a moraic metre, for which you need to establish a pattern based on mora counts
- to look up the name of a metre, refer first to the DHARMA authority file on prosody and verse forms, https://dharmalekha.info/prosody
  - if the metre is not listed in the authority file, consult one of the reference works cited there, in particular
    - Appendix A in Apte's (1957) *Practical Sanskrit Dictionary*; and when this does not help,
    - Velankar's (1949) *Jayadāman* for verse in India
    - Zoetmulder's (1974) *Kalangwan* for verse in Indonesia
- you may also try an online tool that will identify a Sanskritic metre from text that you paste (and clean of markup) or type in a text box, but the identification produced by such a tool should always be verified
  - https://sanskritmetres.appspot.com/ requires a full stanza as input but will tolerate mistakes and lacunae and produce an approximate match
  - http://sanskritlibrary.org:8080/MeterIdentification/ recognises a very large number of metres and accepts half or quarter stanzas as input (but does not tolerate errors)
  - https://www.skrutable.info/ requires full stanzas but can automatically split these into quarters and tolerates mistakes to some extent; it can identify moraic metres in addition to syllabic ones and includes sound recordings of the recitation of many metres

2.2. Divergence from prosodic templates

When a particular instantiation of a metre diverges from what the prosodic template for that metre requires, this may or may not be perceived as bad practice on the part of the composer and may require different encoding approaches, as discussed in the EGD and EGC.

2.2.1. Legitimate variation

2.2.1.1. Lenient templates

- some conventional metres, especially those of pre-classical Sanskrit poetry, use lenient templates, where
  - no length constraint is applied to some of the syllables, as in
    - the first four syllables of *anuṣṭubh*, §2.3.1
    - the first syllable of *upajāti* and *vaṁśamālā*, §2.3.2

    - all eight syllables of the Javanese *pādānuṣṭubh*
  - the number of syllables may vary somewhat, as in pre-classical trimeter, §2.3.3
- this kind of leniency is already represented in our definitions of the relevant metres, so stanzas (or lines) encoded with one of these metres need no additional encoding for their specific instantiation

2.2.1.2. Alternative templates

- some conventional metres admit specific alternative realisations with their own set of constraints, as in
  - *vipulā anuṣṭubh*, §2.3.1
    - these variants are listed in our authority file, so *anuṣṭubh* lines following a *vipulā* pattern may be encoded explicitly as such
  - *vipulā āryā*, §2.4.1
    - these variants differ from the standard *āryā* only in the placement of the caesura and thus require no explicit encoding, but the expected and unobserved caesura may be encoded

2.2.2. Poetic licence

- some schools of versification permit certain kinds of licence (a relaxation of the mechanical rules) in determining syllable length
  - the most common licence found in the Sanskritic tradition is that a short vowel followed by a voiceless stop and an *r* or *l* may count as a short syllable
  - in early epic *anuṣṭubh*, three successive short syllables may sometimes be present in place of two syllables of the first (unconstrained) half of lines
- poetic licence cannot be objectively distinguished from faulty prosody, so it is encoded in the same manner: the expected metre is encoded for the stanza, and the actual instantiation may be encoded for any divergent line
  - this encoding is not a value judgement but a phenomenal description

2.2.3. Deviant prosody

2.2.3.1. Unmetrical verse

- when a line of verse fails to conform to the expected template, it is said to be unmetrical
- unmetrical verse (*chandobhaṅga*, the breaking of the metre) is usually avoided by poets and strongly discouraged by the traditional authorities on versification
  - an oft-cited injunction says, *api māsaṁ maṣaṁ kuryāc chandobhaṅgaṁ na kārayet*, "One may even make a bean into a bin, but must not break the metre"
- a verse line may be unmetrical when one of its constrained syllables does not have the expected length; instead or in addition, it may be
  - hypometrical, when its syllable count (or mora count in moraic verse) is lower than expected
  - hypermetrical, when its syllable count (or mora count in moraic verse) is higher than expected
- while DHARMA practice is always to encode the expected metre for stanzas, unmetrical lines may be optionally furnished with additional encoding to record their actual prosodic instantiation

2.2.3.2. Unobserved caesurae

- in metres whose template includes a caesura, the poet's failure to observe the caesura (*yatibhaṅga*) is generally also considered a grave error on the part of the poet
  - however, some Sanskritic metres require a caesura according to some schools of versification but do not require it according to other schools
- failure to observe an expected caesura may be encoded optionally in DHARMA practice

2.3. Syllabic metres (*varṇavṛtta*)

Regular syllabic metres use a fully constrained template, which may be identical for all four lines of a quatrain (*samavṛtta*), alternate for the odd and even lines (*ardhasamavṛtta*), or be different for each line (*viṣamavṛtta*). While the first two of these kinds make up a large proportion of epigraphic verse, their identification is straightforward and needs no discussion beyond a listing of the templates in the authority file. The following subsections discuss metres in which the template is subject to a limited degree of variation.

2.3.1. *Anuṣṭubh*

In our usage, *anuṣṭubh* is an umbrella term: we do not make a distinction between Vedic *anuṣṭubh*, epic *anuṣṭubh/śloka* and classical *anuṣṭubh/vaktra/śloka*. An *anuṣṭubh* stanza consists of four lines of eight syllables. The first four syllables of each line are largely unconstrained, while the second eight form a metrical cadence which takes one form in the odd lines of the stanza and another form in the even lines (so *anuṣṭubh* is an *ardhasama* metre). The variations of *anuṣṭubh* are shown in Table 3 below, where white cells contain invariant pattern components, yellow shading indicates the distinctive segment of each pattern, and peach shading indicates additional constraints imposed on certain variants. For an in-depth study of this metre, see Jacobi (1885).

- in any line, the second and third syllables must not be both short
  - the first half of a line is thus not completely unconstrained, patterns matching ⏑⏑⏑⏑ being forbidden
- in the default or *pathyā* form of the *anuṣṭubh*,
  - the pattern ⏑––⏑ ends odd lines (a and c, the first and third line)
  - the pattern ⏑–⏑⏑ ends even lines (b and d, the second and fourth line)
- legitimate *anuṣṭubh* variants called *vipulā* have alternative cadences in the odd quarters, as presented in the lower part of Table 3
  - *vipulā* forms are named after the *gaṇa* code (§1.5) of their distinctive cadence, i.e. the first three syllables of the second half
  - the *bha* and *ma* and (probably) the *ta vipulā* constrain the first half of odd lines to the pattern ⏑–⏑–
  - the *ma vipulā* requires a caesura after the fifth syllable
  - the *ra vipulā* requires a caesura after the fourth syllable
  - not all *vipulā* forms are recognised as legitimate by all traditional authorities
    - in particular, the *ma* and *sa vipulā* are generally absent in earlier treatises
  - authorities do not agree fully on the additional constraints applicable to *vipulā* forms
  - the *sa* and *ta vipulā* are very rare in attested poetic practice
  - when a rare and/or rarely acknowledged *vipulā* form occurs in a text, it is usually impossible to be certain whether their composer treated these as legitimate or produced a metrically incorrect stanza

Table 3. Permitted patterns in anuṣṭubh

	1	2–4	5	6	7	8
odd lines (a, c): *pathyā*	⏑	– – ⏑ ⏑ – ⏑ – ⏑ ⏑	⏑	–	–	⏑
even lines (b, d): all forms	⏑	– – ⏑ ⏑ – ⏑ – ⏑ ⏑	⏑	–	⏑	⏑
odd lines (a, c): *na-vipulā*	⏑	– – – ⏑ – – – ⏑ –	⏑	⏑	⏑	⏑
odd lines (a, c): *bha-vipulā*	⏑	– ⏑ –	–	⏑	⏑	⏑
odd lines (a, c): *ma-vipulā*	⏑	– ⏑ –	– ||	–	–	⏑
odd lines (a, c): *ra-vipulā*	⏑	– – – ⏑ – – – ⏑ –	|| –	⏑	–	⏑
odd lines (a, c): *ta-vipulā*	⏑	– ⏑ –	–	–	⏑	⏑
odd lines (a, c): *sa-vipulā*	⏑	– – – ⏑ – – – ⏑ –	⏑	⏑	–	⏑

2.3.2. The *upajāti* group

- this group of metres includes 11 and 12-syllable metres which vary in the length of the first syllable and thus give rise to ambiguities concerning classification
  - *upajāti* or *triṣṭubh upajāti*, a free mix of *indravajrā* and *upendravajrā*
  - *vaṁśamālā* or *jagatī upajāti*, a free mix of *indravaṁśā* and *vaṁśastha*
- when every line of a stanza is in one of the "pure" metres (e.g. *indravajrā*), that stanza should normally be classified as that pure metre, whereas stanzas with one or more lines in the other child metre should be classified as the "mixed metre" (e.g. *upajāti*)
- however, the mixed metres are more widely used than the pure ones, therefore
  - if an inscription includes several successive stanzas of a mixed metre among which one or a few stanzas are in a pure metre, then it makes better sense to classify the pure stanza(s) as being also of the mixed metre (assuming that the poet was composing in the mixed metre and by chance all lines of that particular stanza turned out in one of the pure metres)
  - if an inscription includes a stanza in one of these metres with at least one line-initial syllable lost, then it is better to assume the stanza to be in the mixed metre even if all the fully extant lines are in one of the pure metres
  - there may always be cases where the above considerations do not apply; for example when a composer shows off his skill by employing a wide variety of metres

2.3.3. Pre-classical trimeter

The intricacies of partially constrained Vedic metres are beyond the scope of our interest. Some stanzas attested in epigraphy and in the manuscript tradition do, however, conform to pre-classical conventions. As

noted above (§2.2.1), Vedic *anuṣṭubh* is for our purposes merely a metrically deviant version of classical *anuṣṭubh*. Slightly more common in the corpora we deal with are stanzas consisting of eleven- and/or twelve-syllable lines that do not conform to the strict constraints of *indravajrā/upendravajrā* or *indravaṁśā/vaṁśastha*, nor to the minimally relaxed constraints of *upajāti* or *vaṁśamālā* (§2.2.2). We refer to such metres collectively as trimeter following Arnold (1905, 7, 11–14).

- the general characteristics of trimeter verse are the following
  - lines consist of 11±1 or 12±1 syllables, with varying line number permitted within a stanza
  - there is a more or less clear caesura after the fourth or fifth syllable
  - the initial colon (the "opening") is relatively free, but predominantly trochaic
  - the caesura is generally followed by a pair of short syllables (the "break")
  - the final part of each line (the "cadence") is relatively fixed in a trochaic pattern
- we judge that a rough typology of metrical patterns serves our needs better than a detailed encoding that could give due consideration to the intricacies of these metres, and hence we classify them into the following rough categories
  - *triṣṭubh* is our term for stanzas of predominantly 11-syllable lines which predominantly conform to the pattern ⏑–⏑–||⏑⏑––⏑–⏑ or ⏑–⏑–⏑||⏑⏑–⏑–⏑ as shown in Table 4
  - *jagatī* is our term for stanzas of predominantly 12-syllable lines which predominantly conform to the pattern ⏑–⏑–||⏑⏑––⏑–⏑⏑ or ⏑–⏑–⏑||⏑⏑–⏑–⏑⏑ as shown in Table 4
  - *trimeter* is our fallback term for stanzas where the metrical pattern and/or length of the lines varies more than in the more specific metres named above

Table 4. Templates of triṣṭubh *and* jagatī

	opening and break	cadence
triṣṭubh	⏑–⏑–||⏑⏑– (early caesura) or ⏑–⏑–⏑||⏑⏑ (late caesura)	–⏑–⏑
jagatī		–⏑–⏑⏑

2.3.4. The *daṇḍaka* class

The *daṇḍaka* is a rather vaguely defined class of syllabic metres with very long lines. Their common feature is an arbitrary (but usually large) number of iterations of a basic trisyllabic sequence, typically the *ra-gaṇa* (–⏑–), *ya-gaṇa* (⏑––) or the *sa-gaṇa* (⏑⏑–). This iterative part is most often preceded by a fixed sequence of six short syllables, but metres classified as *daṇḍaka* may also lack a fixed sequence, contain a different fixed sequence, or have a fixed sequence at the end rather than at the beginning.

2.4. Moraic metres

The metres in this section, called *āryā* or *gāthā* in the Sanskritic tradition, are based on moraic feet (*gaṇa*), and are therefore also referred to as *gaṇacchandas*. They consist of two hemistichs, each comprised of eight feet and divided by a caesura after the third foot. Indian theorists usually regard the resulting four cola as four quarters of a quatrain, but in our encoding and discussion we treat each hemistich of such a stanza as a line. Foot boundaries must coincide with syllable boundaries (i.e. a long syllable must never be shared between the end of one foot and the beginning of the next foot), but not necessarily with word boundaries. The varieties of *gaṇacchandas* metres are summarised below, with their line structure illustrated in Table 5 (where white cells contain invariant pattern components, yellow shading indicates the distinctive segment of each pattern, and peach shading indicates additional constraints imposed on certain variants), and their stanza structure illustrated in Table 6 and Table 7. For an in-depth study see Jacobi (Jacobi 1886) and Ollett (2013).

2.4.1. The structure of moraic lines

Lines (hemistichs) of the *āryā* class consist of eight feet, the first seven of which are tetramoraic (consist of four morae), while the last is bimoraic (consists of two morae), comprised of a single syllable that is always regarded as long.

- the foot structure permits the following variations, illustrated in Table 5 below
  - 7 × 4 + 2 = 30 morae, the basic line template
  - 5 × 4 + 1 + 4 + 2 =27 morae, a shortened template where the sixth foot is reduced to a single mora (one short syllable)
  - 8 × 4 = 32 morae, a lengthened template where the eighth foot is expanded to four morae
  - some theorists mention a combination of the two variants, i.e. a line of 5 × 4 + 1 + 4 + 4 =29 morae, but such a line is not known to exist in extant literature (Warder 1967, 143)
- beyond the mora count, the following constraints apply to the prosodic composition of the individual feet, also illustrated in Table 5 below
  - in the odd-numbered feet (1, 3, 5 and 7) the pattern ⏑–⏑ is forbidden
    - if the pattern is ⏑⏑⏑⏑ in the fifth foot of a 27-mora line or the seventh foot of any line, then this foot must be preceded by a secondary caesura (i.e. begin at a word beginning)
  - in the even-numbered tetramoraic feet (2, 4 and 6 unless it is reduced) any mora structure is permitted
    - the pattern ⏑–⏑ is often preferred, but not required
    - in foot 6, only the following patterns are permitted
      - either ⏑–⏑
      - or ⏑||⏑⏑⏑, with a secondary caesura after the first syllable
- a line (of any foot structure) that conforms to the above constraints is called a *pathyā* line, but the following alternatives are also available
  - a lifting or shifting of the mandatory caesura after the third foot is called *vipulā*, in which case the fourth foot must have one of the following patterns
    - either ⏑–⏑
    - or ⏑||⏑⏑⏑, with the caesura shifted to the point after this foot's first syllable
  - a stricter constraint imposed on the first five feet (requiring the pattern ⏑⏑–⏑–⏑––||⏑–⏑–⏑⏑) is called a *capalā*
  - stanzas involving one or two *capalā* lines have the following designations
    - *mukhacapalā* if the constraint is applied to the first hemistich
    - *jaghanacapalā* if the constraint is applied to the second hemistich
    - *mahācapalā* or *ubhayacapalā* if the constraint is applied to both hemistichs
  - neither *vipulā* nor *capalā* forms are encoded explicitly in DHARMA practice, but encoders wishing to promote the study of prosody may encode the former as an unobserved caesura and the latter as an actual prosodic realisation

Table 5. Prosodic templates of lines in moraic metres

	1	2	3	4	5	6	7	8
generic templates								
30 morae	⏑⏑ ⏑⏑	⏑–⏑ ⏑⏑ ⏑⏑	⏑⏑ ⏑⏑	‖⏑–⏑ ‖⏑⏑ ⏑⏑	⏑⏑ ⏑⏑	⏑–⏑ ⏑‖⏑⏑⏑	‖⏑⏑⏑⏑ ⏑⏑– –⏑⏑ ––	⏑
27 morae	⏑⏑ ⏑⏑	⏑–⏑ ⏑⏑ ⏑⏑	⏑⏑ ⏑⏑	‖⏑–⏑ ‖⏑⏑ ⏑⏑	‖⏑⏑⏑⏑ ⏑⏑– –⏑⏑ ––	⏑	‖⏑⏑⏑⏑ ⏑⏑– –⏑⏑ ––	⏑
32 morae	⏑⏑ ⏑⏑	⏑–⏑ ⏑⏑ ⏑⏑	⏑⏑ ⏑⏑	‖⏑–⏑ ‖⏑⏑ ⏑⏑	⏑⏑ ⏑⏑	⏑–⏑ ⏑‖⏑⏑⏑	‖⏑⏑⏑⏑ ⏑⏑– –⏑⏑ ––	⏑⏑ ⏑
variations and constraints								
capalā	⏑⏑–	⏑–⏑	––	‖⏑–⏑	–⏑⏑			
vipulā				⏑–⏑ ⏑‖⏑⏑⏑				

2.4.2. The structure and classification of moraic stanzas

The varieties of moraic lines discussed in §2.3.1 above can be used in various combinations, but not all combinations occur in poetic practice, and the names assigned to the individual combinations are not entirely consistent across the schools of the tradition. In DHARMA practice we assign the following names to the actually occurring combinations, shown in list form in Table 6 and crosstabulated in Table 7.

- all possible combinations of 30-mora and 27-mora hemistichs occur widely:
  - the *āryā* of 30/27 morae, the most common combination and the archetypal *gaṇacchandas* metre
  - the *gīti* of 30/30 morae
  - the *udgīti* of 27/30 morae
- there is only one common combination involving the 32-mora line:
  - the *āryāgīti* of 32/32 morae
- other combinations are barely attested in known literature and their nomenclature is rarely and inconsistently given (Warder 1967, 143–44; Cappeller 1872, 14, 21–24; Colebrooke 1811), but since several of these forms do occur in some of our epigraphic texts, we classify and name them as follows
  - *sugīti*: 32/27 morae
  - *anugīti*: 27/32 morae
  - *vallarī*: 32/30 morae
  - *lalitā*: 30/32 morae

Table 6. Names and foot structure of stanzas in moraic metres

	Line structure	Foot structure (detailed template in Table 5)
common combinations		
āryā	30 / 27 morae	4|4|4|4|4|4|4|2/ 4|4|4|4|4|1|4|2
gīti	30 / 30 morae	4|4|4|4|4|4|4|2/ 4|4|4|4|4|4|4|2
upagīti	27 / 27 morae	4|4|4|4|4|1|4|2/ 4|4|4|4|4|1|4|2
udgīti	27 / 30 morae	4|4|4|4|4|1|4|2/ 4|4|4|4|4|1|4|2
āryāgīti	32 / 32 morae	4|4|4|4|4|4|4|4/ 4|4|4|4|4|4|4|4
uncommon combinations		
sugīti	32 / 27 morae	4|4|4|4|4|4|4|4/ 4|4|4|4|4|1|4|2
anugīti	27 / 32 morae	4|4|4|4|4|1|4|2/ 4|4|4|4|4|4|4|4
vallarī	32 / 30 morae	4|4|4|4|4|4|4|4/ 4|4|4|4|4|4|4|2
lalitā	30 / 32 morae	4|4|4|4|4|4|4|2 4|4|4|4|4|4|4|4

Table 7. Names for moraic metres with various hemistich lengths

ab → / cd ↓	27	30	32
27	*upagīti*	*āryā*	*sugīti*
30	*udgīti*	*gīti*	*vallarī*
32	*anugīti*	*lalitā*	*āryāgīti*

2.5. Metres combining syllabic and moraic cola

2.5.1. The *vaitālīya* group

This group of *ardhasama* metres involves a loose moraic template for the first part of each line (the opening) and a syllabic template for the final part of each line (the cadence). All of these metres have an opening of six morae in the odd lines and eight morae in the even lines, followed by a cadence of five or six syllables distinctive of the metre, illustrated in Table 8.

- the following constraints apply to the moraic opening
  - none of the even morae may belong to a long syllable shared with the next mora
    - in other words, the openings may be understood as consisting of bimoraic feet (as shown in Table 8) or as a series of units comprised of one long or two short syllables (as shown in the formulae below)
  - the openings must not consist entirely of long syllables, nor entirely of short syllables

- the cadences and the associated metre names are as follows
  - *vaitālīya* has the cadence –⏑–⏑⏓
    - its template is thus ⏑⏑⏑⏑–⏑–⏑⏓/⏑⏑⏑⏑⏑⏑–⏑–⏑⏓
  - *aupacchandasika* has the cadence –⏑–⏑–⏓
    - its template is thus ⏑⏑⏑⏑–⏑–⏑–⏓/⏑⏑⏑⏑⏑⏑–⏑–⏑–⏓
  - the much rarer *āpātalikā* has the cadence –⏑⏑–⏓
    - its template is thus ⏑⏑⏑⏑–⏑⏑–⏓/⏑⏑⏑⏑⏑⏑–⏑⏑–⏓
- some realisations of the above templates are themselves named and recognised syllabic metres
  - *vaitālīya* may be realised as
    - *viyoginī*: ⏑⏑–⏑⏑–⏑–⏑–⏓/⏑⏑––⏑⏑–⏑–⏑–⏓
    - *aparavaktra*: ⏑⏑⏑⏑⏑⏑–⏑–⏑–⏓/⏑⏑⏑⏑–⏑⏑–⏑–⏑–⏓
  - *aupacchandasika* may be realised as
    - *mālabhāriṇī*: ⏑⏑–⏑⏑–⏑–⏑–⏑–⏓/⏑⏑––⏑⏑–⏑–⏑–⏑–⏓
    - *puṣpitāgrā*: ⏑⏑⏑⏑⏑⏑–⏑–⏑–⏑–⏓/⏑⏑⏑⏑–⏑⏑–⏑–⏑–⏑–⏓
  - *āpātalikā* may be realised as
    - *vegavatī*: ⏑⏑–⏑⏑–⏑⏑–⏑–⏓/–⏑⏑–⏑⏑–⏑⏑–⏑–⏓
  - in actual poetic practice, these fully constrained realisations are much more common than the flexible moraic templates
    - nonetheless, many editors of Indic texts prefer to classify such stanzas by the name of the generic metre and not that of the specific realisation
    - this should be avoided in DHARMA editions: if a previous edition identifies a stanza as one of these generic metres, then instead of blindly following their nomenclature check whether the stanza in fact conforms to one of the specific metres, and if it does, mark it up as such

Table 8. Metres of the vaitālīya *group*

metre	lines	opening	cadence
vaitālīya	odd	2|2|2|	–⏑–⏑⏓
	even	2|2|2|2|	
aupacchandasika	odd	2|2|2|	–⏑–⏑–⏓
	even	2|2|2|2|	
āpātalikā	odd	2|2|2|	–⏑⏑–⏓
	even	2|2|2|2|	

2.5.2. *Gītikā*

A rare metre not discussed in the standard treatises on versification but found in some epigraphic texts, provisionally named *gītikā* (Venkatasubbiah 1935), consists of four lines of length varying between 9 and 13 syllables, exhibiting a different pattern in the odd and even lines. The cadence of the lines is similar to that of the *anuṣṭubh* except that two short syllables may replace the penultimate long syllable in odd lines.

- based on the known attestations (some of which are damaged), the following generic template may be established
  - odd lines: ⏑⏑–⏓⏑⏑–⏓⏑⏑⏓ (where the sequence ⏓⏑⏑ usually manifests as –⏑ and never as ⏑⏑⏑)
  - even lines: ⏑⏑–⏑––⏑–⏑⏓ (possibly with ⏑⏑ permitted in place of the first –)

3. The Tamil versification tradition

This section has been contributed by Emmanuel Francis. Due to practical considerations, Tamil metre is presented here as a brief summary based on Niklas 1988.

- Tamil metre is classified by major types (*pā*) and their forms and subtypes (*pāviṉam*) as shown in Table 9 below
- the names in any column may be used as values of `@met` in `<lg>`
- a metrical line (*aṭi*) is made up of a variable number of metrical feet (*cīr*), from 2 to more than 5
- a metrical foot (*cīr*) is made of a variable number of metremes (*acai*), from 1 to 4
- there are two main kinds of metremes (*acai*):
  - the monosyllabic *nēr-acai*
  - and the disyllabic *nirai-acai*
- in ancient prosody there are also *nērpu* and *niraipu*, that is, one *nēr* or *nirai* followed by a syllable containing an overshort *u* or normal short *u*
- a metreme (*acai*) is made of a variable number of letters or phones (*eḻuttu*), from 1 to 3

Table 9. Tamil metres

Major type	Included forms	Included subtypes (*pāviṉam*)
veṇpā	kuṟaḷ-veṇpā (2 *aṭi*s) nēricai-veṇpā (4 *aṭi*s) iṉṉicai-veṇpā (4 *aṭi*s) cintiyal-veṇpā (3 *aṭi*s) paḵṟoṭai-veṇpā (5 to 12 *aṭi*s)	kuṟaḷ-veṇcenturai kuṟaḷ-tāḻicai veṇ-tāḻicai veṇ-turai veḷi-viruttam
āciriyappā (3 to 1000 *aṭi*s)	nēricai-āciriyappā iṇaikkuṟal-āciriyappā nilaimaṇṭila-āciriyappā aṭimaṟimaṇṭila-āciriyappā	āciriya-tāḻicai āciriya-turai āciriya-viruttam
kalippā[2]	ottāḻicai-kalippā (with 3 subforms) veṇ-kalippā koccaka (with 5 subforms, the 5th one having 3 subsubforms)	kali-tāḻicai kali-turai kali-viruttam kaṭṭalai-kalitturai kaṭṭalai-kalippā
vañcippā[3]	(no forms)	vañci-tāḻicai vañci-turai vañci-viruttam
maruṭpā	composed of elements of veṇpā and āciriyappā	

[2] Contains up to 6 different types of elements which are: *taravu*, *tāḻicai*, *arākam*, *ampōtaraṅkam*, *taṉiccol*, *curitakam*.

[3] Contains elements which are: *taṉiccol*, *akaval-curitakam*.

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